Econ 527 HW4
Wealth Distribution in Incomplete Market Models

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1 The Huggett Model

The equilibrium interest rate is defined as

$$r^* := A(r^*) = 0,$$

i.e. net asset demand is zero at the equilibrium rate. For this model with the given parameters, $r^* = -0.0135$. The equilibrium was found using a bracketing algorithm similar to bisection, with bounds $r_{\text{low}} = -0.05$ and $r_{\text{high}} = \frac{1}{\beta} - 1$. The top of the wealth grid is not binding, but some households do hit the borrowing limit.

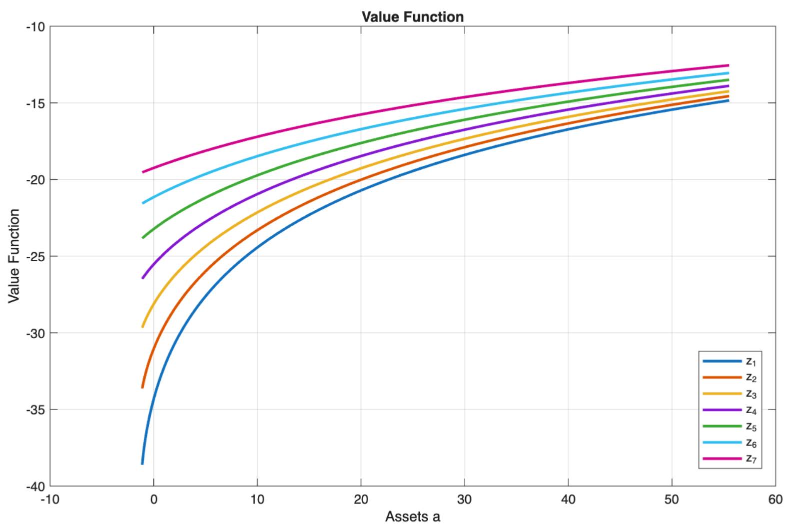

Figure 1: Value Function — Huggett

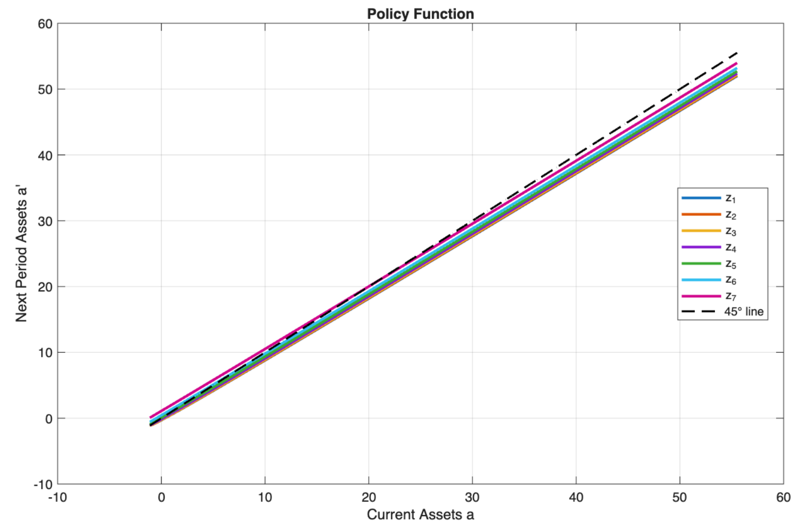

Figure 2: Policy Function — Huggett

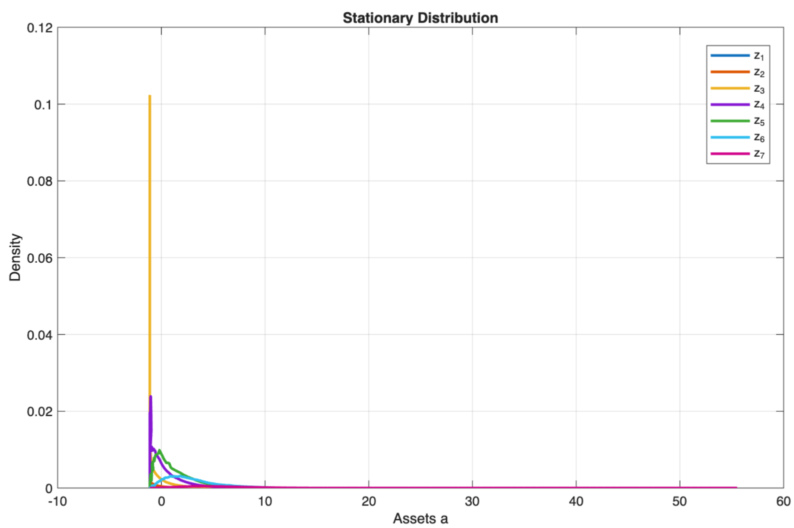

Figure 3: Stationary Distribution — Huggett

2 The Aiyagari Model

The equilibrium condition in this model is

$$r^* := S(r^*) = K(r^*),$$

i.e. household asset supply equals firm capital demand. The equilibrium interest rate was found to be $r^* = 0.023$ via fixed-point iteration on the capital market clearing condition.

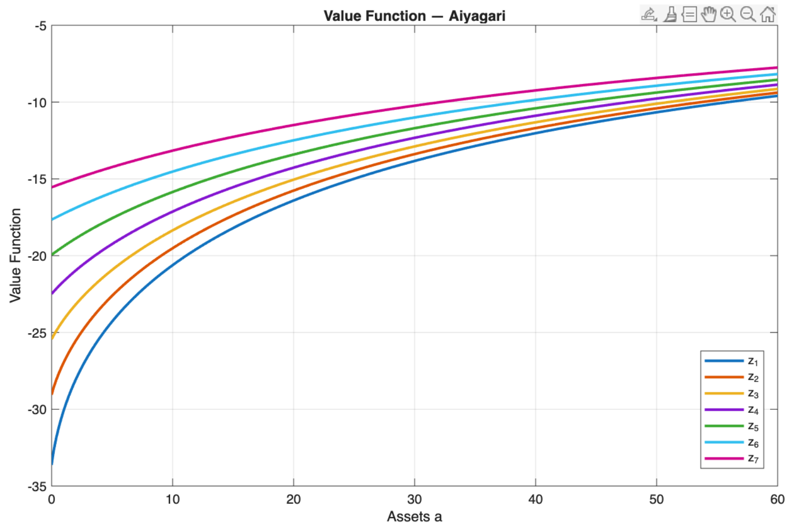

Figure 4: Value Function — Aiyagari

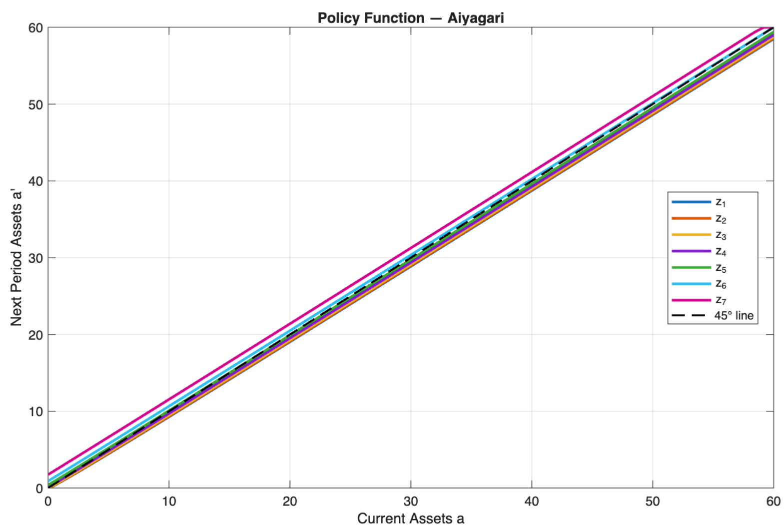

Figure 5: Policy Function — Aiyagari

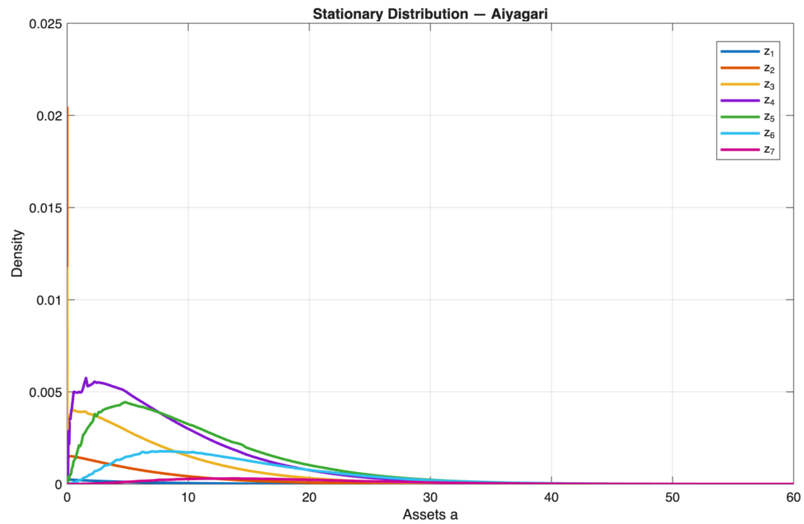

Figure 6: Stationary Distribution — Aiyagari

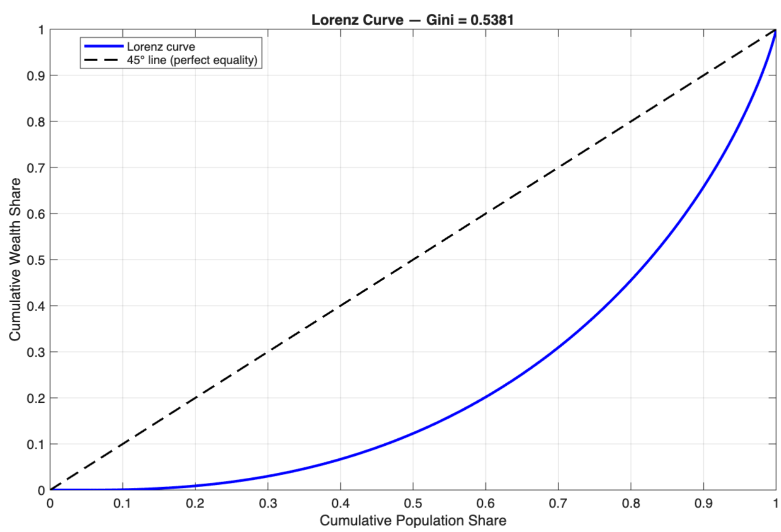

Figure 7: Lorenz Curve — Aiyagari

Table 1 reports the Gini coefficient and top wealth shares for the Aiyagari model alongside the 2019 Survey of Consumer Finances (SCF). The model generates substantially less wealth inequality than is observed empirically: the Gini coefficient is 0.54 compared to 0.85 in the data, and the top 10% and 1% shares fall well short of their empirical counterparts. This is a well-known limitation of the standard Aiyagari model, which relies solely on precautionary savings to generate wealth dispersion. We do not compute inequality statistics for the Huggett model because asset positions in that economy are net bond holdings, which can be negative, making comparisons to SCF wealth estimates difficult to interpret.

Table 1: Wealth Inequality: Model vs. Data

Statistic	2019 SCF	Aiyagari Model
Gini coefficient	0.85	0.5381
Top 10% share	64.6%	34.2%
Top 1% share	29.6%	5.5%

3 Net Asset Demands

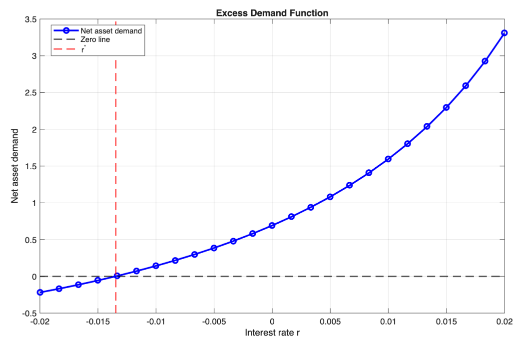

Figure 8: Excess Demand — Huggett

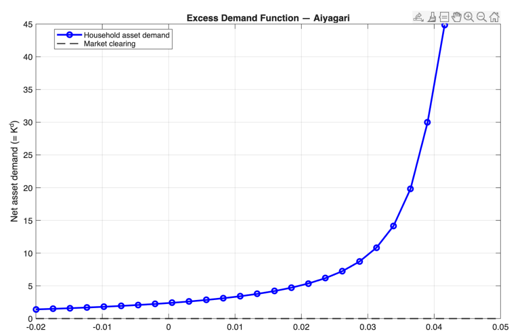

Figure 9: Excess Demand — Aiyagari

4 Code Description

The model is solved using the following MATLAB files. The two main scripts call the shared utility functions and solvers listed below.

5 AI Usage

Claude (Anthropic) was used to debug MATLAB code and to assist with formatting tables and figures in LaTeX.

Table 2: Summary of Code Files

File	Description
`PS4_huggett.m`	Main script for the Huggett model. Calibrates parameters, constructs grids, finds the equilibrium interest rate r^* via `fminbnd`, and plots the value function, policy function, and stationary distribution.
`PS4_aiyagari.m`	Main script for the Aiyagari model. Solves for the general equilibrium capital stock K^* via damped iteration on the capital market clearing condition, and computes wealth inequality statistics (Gini, top 10% and 1% shares).
`f_spe.m`	Solves the household stationary partial equilibrium given an interest rate r and income process. Implements the Endogenous Grid Method (EGM) for the value and policy functions, constructs the transition matrix, and returns the invariant distribution and net asset demand.
`f_nad.m`	Wrapper around `f_spe` that returns only net asset demand. Used as the objective function in root-finding and for plotting the excess demand curve.
`rouwenhorst.m`	Discretizes an AR(1) income process into an n_z-point grid and transition matrix using the Rouwenhorst method.
`mc_invdist.m`	Computes the invariant distribution of a Markov chain via eigendecomposition of the transition matrix.
`polynomial_grid.m`	Constructs a curved asset grid on $[a_{\min}, a_{\max}]$ using a polynomial transformation, concentrating grid points near the borrowing constraint.